
BARCELONA COMPUTATIONAL FOUNDATION

Working Paper WP0125

Slide Deck · Pillar: AI Agents / Kolmogorov Theory

Agent Souper

A liquid agent-system substrate for telehomeostasis and meta-agents (Beta)

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What is Agent Souper?

An open, local-first simulation platform — a working **Beta** — for **liquid agent systems**: populations of Kolmogorov-Theory agents in a shared 2D world, coupled by a dynamical connectome.

- Executable counterpart of the *Agent Soups* formalism; built to the L1/L2/L3 agent taxonomy.
- Every L3 agent is an *explicit* KT decomposition — the objective function is always external (never fused into a net).
- Three configurable worlds on one engine: Simple Demo (L1 control), Predator–Prey (L3), Soccer (L3 possession game).
- Purpose: make two KT questions empirical — **telehomeostasis** and the emergence of **meta-agents**.

Remark: Design discipline

Objective function external · deterministic fast loop the LLM may only *modulate* · no collective orchestrator · substrate-tested experiments. This is what keeps the science readable.

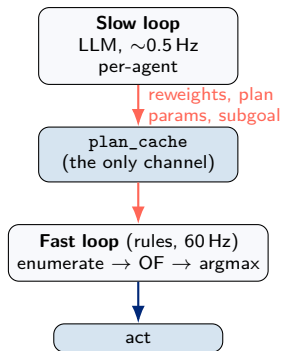
The KT agent, and the dual-rate contract

L3 = persona + 7 modules + membrane

- world model · self model
- **objective function** (external)
- simulator · comparator
- updater · planner

Fast loop, every tick:

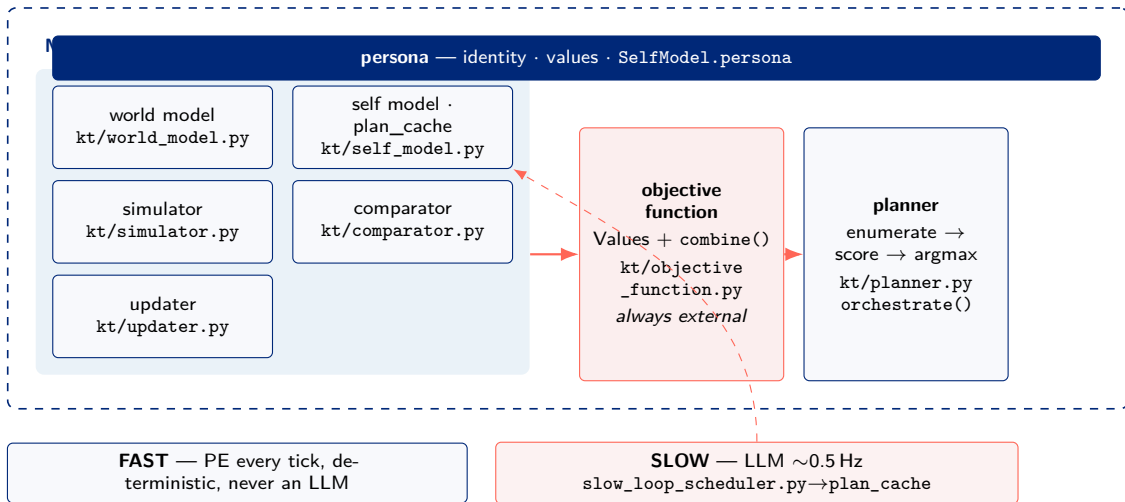
$$a^* = \arg \max_a \frac{\sum_v w_v s_v(a)}{\sum_v w_v}, \quad w_v = 0.5^{p_v - 1} r_v$$



The engine *never* awaits the LLM. Off $\Rightarrow$ exact rules baseline, persona inert.

The agent model, mapped to the software

MEMBRANE — MCP-as-contract



Telehomeostasis — without a conductor

Definition: Telehomeostasis (operational)

Each member carries an objective whose argument is the persistence of the *kind* (lineage, team), and acts on it from *local* information only. A compressed collective objective — not a directive from a controller.

- **No collective orchestrator.** The LLM is bound to one agent and sees only what that agent sees — as in nature.
- Instantiated locally: (a) persona defines success collectively; (b) the OF stack serves the kind (e.g. `protect_kin` draw-fire, `team_play` pass-first); (c) a *shared observable* (the scoreboard) each agent perceives.
- The score is observation-only — it never enters the fast OF/valence path. Coordination must *emerge*.

What the soup shows — reported honestly

Soccer (EXP-001)

Telehomeostatic team-play beats an identical selfish Sacrificial draw-fire prey persist $0.57\times$ selfish, 0/8 — objective $7/8$ seeds, 23–2 ($\sim 11\times$).

But the sign flipped through *five* substrate revisions Altruism's value is contingent on the environment's — only an honest substrate (real kickoff) keeps it. binding constraint.

Predator–Prey (EXP-002)

hypothesis **refuted**.

Remark: The methodological result

Do not believe a collective-objective effect until the substrate can no longer flatter it. The refutation and the substrate-dependence are *results*, not failures.

Hypothesis: Meta-agent emergence

A collective behaves as a KT meta-agent when its members each internalise a shared objective, observe a shared signal, and reason locally — with *no* central orchestrator. Meta-agency is then to be *measured*, not imposed.

- A telehomeostatic team is a candidate meta-agent: shared goal, shared observable, local reasoning, emergent competence (EXP-001) an identical crowd cannot produce.
- Open programme: estimate, on the live soup, the collective's model compression and its algorithmic mutual information with its own future — the agenthood threshold.
- Roadmap: collective-agenthood metric · forward-model self-tuning · networked-MCP lift · richer object/field world.

- Ruffini & Castaldo. *Agent Soups: Multiscale Persistence and the Dynamical Connectome*. BCOM, 2026.
- Ruffini. *Layered Persistence: Generalizing Life as Telehomeostatic Agency Across Natural and Artificial Substrates*. BCOM, 2026.
- Ruffini. *Kolmogorov Theory: State of the Art Overview, 2000–2026*. BCOM, 2026.
- Ruffini with Claude Opus 4.7. *Agent Souper* (full treatment). BCOM Working Paper WP0125, 2026.